

H, Q_{in}



Surge Tank

State: H_s (or V_s)

Parameter: A_s

Mass balance:

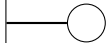
$$A_s \dot{H}_s = Q_{\text{in}} - Q_{\text{out}}$$

Hydrostatic relation:

$$p_s = p_{\text{atm}} + \rho g H_s$$

Optional storage form:

$$\dot{V}_s = Q_{\text{in}} - Q_{\text{out}}$$



H_s, Q_{out}